

Solution Requirements

Fit Buddy AI – AI Fitness Plan Generator using Gemini Models

Generative AI Project | FastAPI + Google Gemini

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Functional requirements

- Accept name, user ID, age, weight, fitness goal and workout intensity
- Generate a structured 7-day plan with warm-up, workout, cooldown and recovery tip for each day
- Provide a nutrition / recovery tip based on the goal
- Update the plan using user feedback
- Store users and plans in a SQLite database
- Show all users and plans in an admin view
- Provide JSON API endpoints with Swagger documentation

Non-functional requirements

- Simple, responsive interface
- Input validation (age 13–100, weight 20–400 kg, intensity low / medium / high)
- Automatic fallback when Gemini is unavailable
- Maintainable Python code
- Secure handling of API credentials
- General wellness disclaimer – not medical advice